

Experiment 02

Advanced SQL Queries

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Aim

To study and implement advanced SQL queries using conditional logic, aggregate functions, joins, subqueries, self-joins, and date operations to solve real-world database problems.

Question 1

A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments.

Answer

```

1 SELECT
2     d.department_name ,
3     ROUND(AVG(g.marks),2) AS average_marks
4 FROM students s
5 JOIN departments d
6 ON s.department_id = d.department_id
7 JOIN grades g
8 ON s.student_id = g.student_id
9 JOIN subjects sub
10 ON g.subject_id = sub.subject_id
11 GROUP BY d.department_name
12 ORDER BY average_marks DESC;
```

Output

	dept_name character varying (50) 	Average Marks numeric 
1	Mathematics	95.00
2	Computer Science	83.33

Question 2

Find the highest paid employee in every department. If multiple employees have the same highest salary, display all of them.

Answer

```

1 SELECT
2     d.dept_name ,
3     e.name ,
4     e.salary
5 FROM employee_02 e
6 JOIN department_02 d
7 ON e.department_id = d.id
8 WHERE (e.department_id ,e.salary)      IN
9 (
10     SELECT
11         department_id ,
12         MAX(salary)
13     FROM employee_02
14     GROUP BY department_id
15 )
16 )
17 ORDER BY d.dept_name;
```

Output

	dept_name character varying (50) 🔒	name character varying (50) 🔒	salary integer 🔒
1	IT	JIM	90000
2	IT	MAX	90000
3	SALES	HENRY	80000

Question 3

Part B: Find the second highest earning employee from the same schema.

Answer

```

1 SELECT
2     d.dept_name ,
3     e.name ,
4     e.salary
5 FROM employee_02 e
6 JOIN department_02 d
```

```

7  ON e. department_id    = d.id
8  WHERE e.salary        =
9  (
10     SELECT MAX(salary)
11     FROM employee_02
12     WHERE department_id = e. department_id
13     AND salary <
14     (
15         SELECT MAX(salary)
16         FROM employee_02
17         WHERE department_id = e. department_id
18     )
19 )
20 ORDER BY d. dept_name;

```

Output

	dept_name character varying (50) 🔒	name character varying (50) 🔒	salary integer 🔒
1	IT	JOE	70000
2	SALES	SAM	60000

Question 4

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to merge these datasets and identify each unique employee (by EmpID) along with their lowest recorded salary across both systems.

Objective:

- Combine the records from Table A and Table B.
- Return each EmpID with the corresponding Employee Name and the lowest salary recorded across both tables.

Answer

```

1  SELECT
2      EmpID ,
3      Ename ,
4      MIN(Salary) AS Lowest_Salary
5  FROM
6  (
7      SELECT EmpID , Ename , Salary
8      FROM TableA
9
10     UNION ALL
11

```

```

12      SELECT EmpID , Ename , Salary
13      FROM TableB
14  ) AS Combined
15
16  GROUP BY
17      EmpID ,
18      Ename
19
20  ORDER BY
21      EmpID;
  
```

Output

	empid integer	ename character varying (50)	lowest_salary integer
1	1	AA	1000
2	2	BB	300
3	3	CC	100

Learning Outcome

- Learned to use aggregate functions for data analysis.
- Applied JOIN operations on multiple related tables.
- Implemented GROUP BY and AVG functions.
- Retrieved highest and second-highest records using SQL.
- Improved PostgreSQL query writing skills.